WEEK 2 · LESSON 9 OF 24 · HANDOUT

Supervised Learning

Study time: **about 50 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=W01tIRP_Rqs — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50 AI, Lecture 4: Learning (supervised, reinforcement, unsupervised) — https://www.youtube.com/watch?v=E4M_IQG0d9g

Your goal in this lesson: Explain how supervised learning works, describe four common supervised methods in plain words, and choose one for a simple problem.

Learning with a teacher

In **supervised learning**, every training example comes with the **correct answer** (the label). The learning algorithm acts like a student who sees many solved exercises, finds the pattern, and then answers new exercises alone.

The process:

1. Show the model many examples of features **and** labels.
2. The model makes predictions and measures how wrong they are (the **error** or **loss**).
3. It adjusts itself to reduce that error.
4. Repeat until the error stops falling.
5. Use the finished model on new examples that have no label.

Supervised learning handles both **classification** (category answers) and **regression** (number answers).

Four common methods

1. Linear regression — draws the best straight line through the data. *Example:* house price rises by about a fixed amount per extra square metre. Simple, fast and easy to explain.

2. Logistic regression — despite its name, it is used for **classification**. It outputs a probability between 0 and 1, such as "82% chance this transaction is fraud". Widely used in banking and health because it is easy to explain.

3. Decision tree — asks a series of yes/no questions, like a flowchart: *Monthly sales above 100,000 RWF? → Late payments fewer than 2? → Likely to repay.* Easy for people to read. A **random forest** combines many trees and is usually more accurate.

4. k-nearest neighbours (k-NN) — to classify a new example, look at the *k* most similar past examples and take a vote. "Members who look like you mostly repaid, so you probably will too."

Method	Problem type	Strength	Weakness
Linear regression	Regression	Simple, explainable	Only straight-line patterns
Logistic regression	Classification	Gives probabilities, explainable	Simple patterns only
Decision tree / random forest	Both	Readable rules (tree), strong accuracy (forest)	One tree can overfit
k-NN	Both	No training step, intuitive	Slow on large data; needs scaled features

Neural networks (Week 3) are also supervised learners — the most powerful kind for images, sound and language.

Where the labels come from

Labels are often the expensive part. Someone must mark thousands of leaf photos as *healthy* or *diseased*, or emails as *spam* or *not spam*. This is called **annotation** or **labelling**, and it is real work done by people — including many data workers in Africa. Labels must be **accurate and consistent**, or the model learns mistakes.

Real supervised systems around you

- Spam filters (classification)
- Credit scoring (classification)
- Crop-disease detection from photos (classification)
- Delivery-time estimates (regression)
- Electricity demand forecasting (regression)

Remember this: Supervised learning learns from examples with correct answers. Choose linear or logistic regression for simple, explainable models; trees and forests for readable or strong models; neural networks for complex data like images and text.

Name: _____ Date: _____

Activity

Task 1

Draw a small decision tree (3–4 questions) that decides if a customer should get a 10% discount at a shop.

Task 2

Test it on three imaginary customers and write the result for each.

Task 3

Write one sentence on what labelled data you would need to *learn* this tree automatically instead of writing it by hand.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What makes learning 'supervised'?

   A. No data is used

   B. Every training example comes with the correct answer (label)

   C. A manager watches the computer

   D. The model has no errors

2. Logistic regression is mainly used for…

A. Classification, giving a probability
B. Storing data
C. Clustering without labels
D. Drawing pictures

3. A model asks 'Sales above 100,000 RWF? Late payments fewer than 2?' and then decides. This is a…

A. k-means clustering
B. Transformer
C. Linear regression
D. Decision tree

4. How does k-nearest neighbours classify a new example?

A. It asks a person
B. It deletes outliers
C. It draws a straight line
D. It looks at the most similar past examples and takes a vote

5. What is a random forest?

A. A type of clustering
B. A data-cleaning method
C. A dataset about trees
D. Many decision trees combined to give a stronger prediction

6. Why is labelling data important work?

A. Wrong or inconsistent labels teach the model mistakes
B. Labels make files smaller
C. Models do not use labels
D. Labels are only for decoration

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